STAT6611

- Lecture 14 (T, Oct 8)
- solutions posted to Homework 2
- Homework 3 assigned due F, Oct 16 by midnight
- Take home midterm to be given on R, Oct 15 & due F, Oct 23 (one week)
- We'll cancel class on R, Oct 15
- I'll fix our canvas front-page

3 Multivariate Normal Distributions; Noncentral χ^2 and F Distributions; Distributions of Quadratic Forms

3.1 Multivariate Normal Distributions

Definition. Let $\underline{x}$ represent an $n \times 1$ random vector. The moment generating function (mgf) of $\underline{x}$, denoted by $M_{\underline{x}}(\cdot)$, is defined by

rv $\underline{x} \in \mathbb{R}^n$
fixed $t \in \mathbb{R}^n$

$$M_{\underline{x}}(\underline{t}) = E\left[\exp(\underline{t}'\underline{x})\right] = E\left[\exp\left(\sum_{i=1}^{n} x_i t_i\right)\right]$$

if the expectation exists for $-h < t_i < h$ $(i = 1, \ldots, n)$ for some number $h > 0$, where $\underline{t} = (t_1, \ldots, t_n)'$. If this expectation does not exist, then $\underline{x}$ has no mgf.

mgf's are important tools for establishing distributions

Theorem 3.1. *Consider two $n \times 1$ random vectors $\underline{x}_1$ and $\underline{x}_2$, and assume that the mgf of each exists. The cumulative distribution functions (cdf's) of $\underline{x}_1$ and $\underline{x}_2$ are identical if and only if the mgf's are equal for all values of $\underline{t}$ in an open rectangle that includes the origin.*

$\underline{x}_1 \overset{d}{=} \underline{x}_2 \iff M_{\underline{x}_1}(t) = M_{\underline{x}_2}(t)$

($\overset{d}{=}$ equal in distribution)

for all $\underline{t} \in \mathbb{R}$ in some open rectangle/set around $\underline{0} \in \mathbb{R}^n$

Theorem 3.2. *If random vectors* $\underline{x}, \underline{x}_1, \ldots, \underline{x}_m$, *where* $\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}$, *have mgf's denoted by* $M_{\underline{x}}(\cdot)$, $M_{\underline{x}_1}(\cdot), \ldots, M_{\underline{x}_m}(\cdot)$, *then* $\underline{x}_1, \ldots, \underline{x}_m$ *are mutually independent if and only if the mgf's satisfy*

$$M_{\underline{x}}(\underline{t}) = M_{\underline{x}_1}(\underline{t}_1) M_{\underline{x}_2}(\underline{t}_2) \cdots M_{\underline{x}_m}(\underline{t}_m) = \prod_{i=1}^{m} M_{\underline{x}_i}(\underline{t}_i) \qquad \text{for all } \underline{t} = \begin{bmatrix} \underline{t}_1 \\ \vdots \\ \underline{t}_m \end{bmatrix}$$

↑ mgf of $(\underline{x}_1, \ldots \underline{x}_m)$ ↑ product of marginal mgfs

in an open rectangle that includes the origin.

$\Rightarrow$ if $\underline{x}_1, \ldots, \underline{x}_m$ independent, then

$$M_{\underline{x}}(\underline{t}) = E\left[\exp\left(\sum_{i=1}^{m} \underline{t}_i' \underline{x}_i\right)\right] = E\left[\prod_{i=1}^{m} \exp(\underline{t}_i' \underline{x}_i)\right]$$

↑ exponential properties

$$= \prod_{i=1}^{m} E[\exp(\underline{t}_i' \underline{x}_i)] = \prod_{i=1}^{m} M_{\underline{x}_i}(\underline{t}_i)$$

↑ independent; $M_{\underline{x}_i}(\underline{t}_i)$

Theorem 3.3. *Let* A *and* B *be* $n \times n$ *matrices of constants,* $\underline{a}$ *and* $\underline{b}$ *be* $n \times 1$ *vectors of constants,* a_0 *and* b_0 *be constants, and* $\underline{x} = (x_1, \ldots, x_n)'$ *be a vector of variables. If* A *is symmetric and* B *is positive definite, then*

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} (\underline{x}' A \underline{x} + \underline{x}' \underline{a} + a_0) \exp\left[-(\underline{x}' B \underline{x} + \underline{x}' \underline{b} + b_0)\right] dx_1 \cdots dx_n$$

$$= \left(\frac{1}{2}\right) \pi^{n/2} |B|^{-1/2} \left[\operatorname{tr}(AB^{-1}) - \underline{b}' B^{-1} \underline{a} + \left(\frac{1}{2}\right) \underline{b}' B^{-1} A B^{-1} \underline{b} + 2a_0\right] \exp\left[\left(\frac{1}{4}\right) \underline{b}' B^{-1} \underline{b} - b_0\right].$$

Theorem 3.3 is just an integration result that can be useful later.

Definition. A random variable z is said to have a univariate standard normal distribution if the probability density function (pdf) of z is given by

$$n(z\colon 0,1) = (2\pi)^{-1/2} \exp\left(-z^2/2\right), \quad z \in \mathbb{R}$$

or if, equivalently, the mgf of z is given by

$$M_z(t) = \exp\left(t^2/2\right), \quad t \in \mathbb{R}.$$

$$\frac{M_z(t)}{\exp(t^2/2)} = \frac{E\exp(tz)}{\exp(t^2/2)} = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(tz)\exp(-z^2/2)\cdot \exp(-t^2/2)\,dz = 1$$

$$\exp(tz - z^2/2 - t^2/2)$$

$$\exp(-\tfrac{1}{2}(z-t)^2)$$

If random variables $z_1, \ldots, z_k$ are mutually independent and each has a univariate standard normal distribution, then the joint pdf of $\underline{z} = (z_1, \ldots, z_k)'$ is given by

$$n(\underline{z}\colon \underline{0}, I) = (2\pi)^{-k/2} \exp\left[-\left(\frac{1}{2}\right)\underline{z}'\underline{z}\right], \quad \underline{z} \in R^k.$$

joint density

$$= \prod_{i=1}^{k} n(z_i; 0, 1)$$

$$= \prod_{i=1}^{k} \frac{1}{\sqrt{2\pi}} \exp(-z_i^2/2) = (2\pi)^{-k/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{k} z_i^2\right)$$

$\underline{z}'\underline{z}$ sum

The mgf of $\underline{z} = (z_1, \ldots, z_k)'$ is given by

$$M_{\underline{z}}(\underline{t}) = \prod_{i=1}^{k} M_{z_i}(t_i) = \exp\left(\frac{1}{2}\sum_{i=1}^{k} t_i^2\right) = \exp\left[\left(\frac{1}{2}\right)\underline{t}'\underline{t}\right] \qquad \text{for all } \underline{t} = \begin{bmatrix} t_1 \\ \vdots \\ t_k \end{bmatrix} \in \mathbb{R}^k$$

by independence

$\exp(\frac{1}{2}t_i^2)$

Definition. A $k \times 1$ random vector $\underline{z}$ is said to have a $\underline{k}$-variate standard normal distribution if the pdf of $\underline{z}$ is $n(\underline{z} : \underline{0}, I)$ or equivalently if mgf of $\underline{z}$ is $M_{\underline{z}}(\underline{t}) = \exp\left[\left(\frac{1}{2}\right)\underline{t}'\underline{t}\right]$, $\underline{t} \in \mathbb{R}^k$.

↑↑ Var($\underline{z}$)
E($\underline{z}$)

Important observation:

(eg. Σ is like variance matrix)

Let $\underline{\mu} \in \mathbb{R}$ & let $p \times p$ matrix Σ be nonnegative definite with rank$(\Sigma) \geq 1$. $\Sigma \neq 0$

Then, there exists some $k \times p$ matrix Γ where $\Gamma'\Gamma = \Sigma$. ($p \times p$) The value of Γ is not unique and, in fact, k could be taken as p ($k \times p$) or even rank$(\Sigma) \geq 1$.

[since $\Sigma = Q'\begin{bmatrix} D_k & 0 \\ 0 & 0 \end{bmatrix} Q$ holds for orthogonal Q and D_k is $k \times k$ positive diagonal matrix]

Define p-dimensional random vector

$$\underline{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix} = \underline{\mu} + \Gamma'\underline{z}, \quad \text{using a } k\text{-variate standard normal } \underline{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_k \end{pmatrix}$$

As a linear transformation of standard normals, $\underline{y}$ will have a mgf, so how does the distribution of $\underline{y}$ depend on the choice of Γ'?

note: ① $\underline{Y}$ is a linear transformation of standard normal $\underline{Z}$

② $E(\underline{Y}) = \underline{\mu}$ & $\text{Var}(\underline{Y}) = \text{Var}(\Gamma'\underline{Z}) = \Gamma'\underbrace{\text{Var}(\underline{Z})}_{I}\Gamma = \Sigma$

Take $\underline{t} \in \mathbb{R}^p$

$$M_{\underline{Y}}(\underline{t}) = E\exp[\underline{t}'\underline{Y}] = E\exp[\underline{t}'(\underline{\mu} + \Gamma'\underline{Z})]$$
$$= \exp(\underline{t}'\underline{\mu})\, E\exp[\underbrace{\underline{t}'\Gamma'}_{\underline{g}'}\underline{Z}]$$

$M_{\underline{Z}}(\underline{g})$ for $\underline{g} = \Gamma\underline{t}$

$$= \exp(\underline{t}'\underline{\mu})\exp\left(\tfrac{1}{2}\underline{g}'\underline{g}\right)$$

$\underline{t}'\Gamma'\Gamma\underline{t} = \underline{t}'\Sigma\underline{t}$

$$= \exp(\underline{t}'\underline{\mu})\exp\left(\tfrac{1}{2}\underline{t}'\Sigma\underline{t}\right)$$

dist. of $\underline{Y} = \underline{\mu} + \Gamma'\underline{Z}$ doesn't depend Γ directly, only need $\Gamma'\Gamma = \Sigma$

Definition. Let $\underline{x}$ be a $p \times 1$ random vector with mean vector $\underline{\mu}$ and variance-covariance matrix Σ, where $\text{rank}(\Sigma) = k \geq 1$. Let Γ be a $k \times p$ matrix such that $\Sigma = \Gamma'\Gamma$. The random vector $\underline{x}$ is said to have a p-variate normal distribution of rank k if the cdf of $\underline{x}$ is the same as that of the random vector $\underline{\mu} + \Gamma'\underline{z}$, where $\underline{z}$ is a $k \times 1$ random vector with the pdf $n(\underline{z}: \underline{0}, I)$; that is, if

$$\underline{x} \stackrel{d}{=} \underline{\mu} + \Gamma'\underline{z}.$$

Remark. Lemma 2.15 provides the existence of a $k \times p$ matrix Γ such that $\Sigma = \Gamma'\Gamma$.

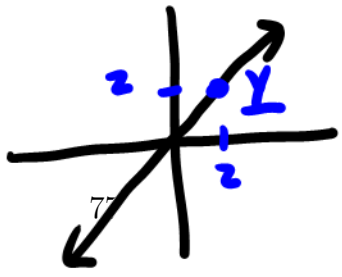
Remark. The definition does not really require that Γ be $k \times p$. Regardless of $\text{rank}(\Sigma) = k$, we may still apply any matrix A where $A'A = \Sigma$ and say $\underline{x}$ has a p-variate normal distribution of rank k if

$$\underline{x} \stackrel{d}{=} \underline{\mu} + A'\underline{z},$$

where A' is $p \times m$ and $\underline{z}$ is a $m \times 1$ random vector with a m-variate standard normal distribution.

Notation. Let $N(\underline{x}: \underline{\mu}, \Sigma)$ denote the cdf of a random vector $\underline{x}$ having a (multivariate) normal distribution with mean vector $\underline{\mu}$ and variance-covariance matrix Σ. We use $\underline{x} \sim N(\underline{\mu}, \Sigma)$ to indicate that $\underline{x}$ is a random vector whose cdf is $N(\underline{x}: \underline{\mu}, \Sigma)$.

Remark. For p-dimensional $\underline{x} \sim N(\underline{\mu}, \Sigma)$, we cannot expect $\underline{x}$ to have a pdf on $\mathbb{R}^p$ unless $\text{rank}(\Sigma) = p$.



Σ^{-1} exists

Theorem 3.4. *Let* $\underline{x} = (x_1, \ldots, x_p)' \sim N(\underline{\mu}, \Sigma)$, *where* $\text{rank}(\Sigma) = p$. *Then,* $\underline{x}$ *has a pdf and it is given by*

$$n(\underline{x}: \underline{\mu}, \Sigma) = (2\pi)^{-p/2}|\Sigma|^{-1/2} \exp\left[-\left(\frac{1}{2}\right)(\underline{x} - \underline{\mu})'\Sigma^{-1}(\underline{x} - \underline{\mu})\right], \quad \underline{x} \in R^p.$$

Theorem 3.5. *Let* $\underline{x} \sim N(\underline{\mu}, \Sigma)$, *where* $\underline{x}$ *is a* $p \times 1$ *vector. Let* B *be a* $q \times p$ *matrix of constants and* $\underline{b}$ *be a* $q \times 1$ *vector of constants. Then,* $\underline{y} = B\underline{x} + \underline{b} \sim N(B\underline{\mu} + \underline{b}, B\Sigma B')$.

$E(Y)$ $Var(Y)$

proof: By def,

$\underline{X} \overset{d}{=} \underline{\mu} + \Gamma'\underline{Z}$ for $\Sigma = \Gamma'\Gamma$ & $\underline{Z} \sim N(\underline{0}, I)$

& then $\underline{Y} = B\underline{X} + \underline{b} \overset{d}{=} B(\underline{\mu} + \Gamma'\underline{Z}) + \underline{b}$

$= B\underline{\mu} + \underline{b} + B\Gamma'\underline{Z}$.

By def,

$\underline{Y}$ is MVN with mean $E(\underline{Y}) = B\underline{\mu} + \underline{b}$ & $Var(\underline{Y}) = B\Gamma'\Gamma B'$

$= B\Sigma B'$

Remark. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector and $\text{rank}(\Sigma) = k \geq 1$. We can retrieve a k-dimensional standard normal vector from $\underline{x}$ as follows.

Take Γ as a $k \times p$ matrix of rank k such that $\Sigma = \Gamma'\Gamma$ and define

$$\underline{y} = (\Gamma\Gamma')^{-1}\Gamma(\underline{x} - \underline{\mu}),$$

which has a k-variate standard normal distribution, because $E(\underline{y}) = \underline{0}$ and

$$\text{Var}(\underline{y}) = (\Gamma\Gamma')^{-1}\Gamma\Sigma\Gamma'(\Gamma\Gamma')^{-1} = (\Gamma\Gamma')^{-1}\Gamma\Gamma'\Gamma\Gamma'(\Gamma\Gamma')^{-1} = I_k.$$

Theorem 3.6. *Suppose that $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Partition $\underline{x}, \underline{\mu}$, and Σ conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \vdots \\ \underline{\mu}_m \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \cdots & \Sigma_{1m} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \cdots & \Sigma_{mm} \end{bmatrix},$$

where $m \geq 2$. Then, $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$ for $i = 1, \ldots, m$.

proof: $\underline{x}_i$ is a linear combination of $\underline{X}$
so $\underline{x}_i$ is $N(\underline{\mu}_i, \Sigma_{ii})$ by theorem 3.5
$\uparrow E(\underline{x}_i)$ $\uparrow \text{Var}(\underline{x}_i)$

Theorem 3.7. *Suppose that* $\underline{x} \sim N(\underline{\mu}, \Sigma)$. *Partition* $\underline{x}, \underline{\mu}$, *and* Σ *conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \vdots \\ \underline{\mu}_m \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \cdots & \Sigma_{1m} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \cdots & \Sigma_{mm} \end{bmatrix},$$

where $m \geq 2$. *Then,* $\underline{x}_1, \ldots, \underline{x}_m$ *are mutually independent if and only if* $\Sigma_{ij} = 0$ *for* $i = 1, \ldots, m$, $j = 1, \ldots, m$, *and* $i \neq j$.

↑ $\text{cov}(\underline{x}_i, \underline{x}_j) = 0$

Proof. For the "if" part ($\Longrightarrow$), suppose that $\Sigma_{ij} = 0$ for $i = 1, \ldots, m$, $j = 1, \ldots, m$, and $i \neq j$. Let $\underline{t} = \begin{bmatrix} \underline{t}_1 \\ \vdots \\ \underline{t}_m \end{bmatrix}$, where $\underline{t}_i$ and $\underline{x}_i$ have the same dimensions for $i = 1, \ldots, m$. Then,

joint mgf

$$M_{\underline{x}}(\underline{t}) = \exp\left[\underline{t}'\underline{\mu} + \left(\frac{1}{2}\right)\underline{t}'\Sigma\underline{t}\right] = \exp\left[\sum_{i=1}^{m}\underline{t}_i'\underline{\mu}_i + \left(\frac{1}{2}\right)\sum_{i=1}^{m}\sum_{j=1}^{m}\underline{t}_i'\Sigma_{ij}\underline{t}_j\right]$$

by def

$\Sigma_{ij} = 0$ $i \neq j$

$$= \exp\left[\sum_{i=1}^{m}\underline{t}_i'\underline{\mu}_i + \left(\frac{1}{2}\right)\sum_{i=1}^{m}\underline{t}_i'\Sigma_{ii}\underline{t}_i\right] = \prod_{i=1}^{m}\exp\left[\underline{t}_i'\underline{\mu}_i + \left(\frac{1}{2}\right)\underline{t}_i'\Sigma_{ii}\underline{t}_i\right] = \prod_{i=1}^{m} M_{\underline{x}_i}(\underline{t}_i),$$

since $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$ by Theorem 3.6.

$M_{\underline{x}_i}(\underline{t}_i)$ since $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$

Since $M_{\underline{x}}(\underline{t})$ is a product,
the $\underline{x}_1, \ldots \underline{x}_m$ are independent

$\Leftarrow$ since $\underline{x}_i, \underline{x}_j$ (have variances) are independent
$\Sigma_{ij} = \text{cov}(\underline{x}_i, \underline{x}_j) = 0$ holds
ie $\text{cov}(\underline{x}_i, \underline{x}_j) = E[(\underline{x}_i - \underline{\mu}_i)(\underline{x}_j - \underline{\mu}_j)']$
ind $= E(\underline{x}_i - \underline{\mu}_i) \cdot E(\underline{x}_j - \underline{\mu}_j)' = 0$
$0 \quad 0$